

FINGAL, Australia

WMO: 959740

Lat: 41.6428S

Lon: 147.9664E

Elev: 778

StdP: 14.29

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	27.9	23.6	18.3	32.7	25.2	19.7	34.8	23.1	52.2	19.2	50.4	1.4	140	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	83.7	62.4	79.7	61.8	76.0	60.6	66.5	75.9	64.7	73.9	63.0	71.6	11.3	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.5	90.4	69.4	61.6	84.3	67.7	59.8	78.9	66.3	31.6	76.1	30.2	73.8	28.9	71.3	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.2	14.2	DB	22.4	93.2	2.1	5.5	20.8	97.1	19.6	100.3	18.4	103.3	16.9	107.3	(4)
(5)			WB	21.7	70.4	2.2	1.6	20.2	71.5	18.9	72.5	17.7	73.4	16.1	74.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	62.2	60.9	58.3	52.8	48.0	44.6	43.8	45.0	48.1	51.4	55.5	58.8		
	DBStd	8.06	5.82	4.99	5.55	5.01	4.75	4.35	3.97	4.10	4.53	5.36	5.52	5.38		
	HDD50	797	0	0	2	26	96	170	196	163	90	43	9	2		
	HDD65	4684	125	133	219	368	527	611	657	619	508	422	289	206		
	CDD50	1679	379	306	261	109	34	8	4	9	32	88	175	274		
	CDD65	91	38	18	13	1	0	0	0	0	2	5	14	14		
	CDH74	1262	547	178	145	12	0	0	0	0	0	38	123	219		
	CDH80	357	196	34	32	1	0	0	0	0	0	4	31	59		
(14) Wind	WSAvg	5.9	7.1	6.3	5.6	4.9	5.3	4.7	5.2	5.8	6.1	6.4	6.7	7.0		
(15) Precipitation	PrecAvg	23.1	1.9	1.4	1.7	2.0	2.0	2.0	2.2	2.5	1.9	1.9	1.9	2.2		
	PrecMax	37.7	7.6	7.8	7.7	7.8	9.7	7.7	6.7	7.5	5.5	5.8	5.9	7.9		
	PrecMin	14.7	0.1	0.0	0.1	0.0	0.1	0.6	0.2	0.7	0.2	0.2	0.4	0.2		
	PrecStd	5.6	1.5	1.5	1.4	1.9	2.0	1.5	1.4	1.6	1.2	1.1	1.1	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	84.9	83.8	75.2	66.6	60.3	59.0	62.7	69.2	79.1	84.3	86.9		
		MCWB	65.4	63.2	62.9	59.4	55.1	52.5	49.8	50.9	54.2	56.7	61.1	64.1		
	2%	DB	84.8	78.9	77.8	70.7	63.2	58.2	56.7	59.3	64.6	71.6	77.1	80.1		
		MCWB	63.3	63.7	61.9	57.6	53.4	51.0	48.9	49.6	52.0	54.4	59.2	60.7		
	5%	DB	79.8	75.3	73.3	66.8	60.7	56.1	54.9	56.9	61.4	67.0	71.8	75.0		
		MCWB	62.0	62.0	60.7	56.2	52.2	49.7	48.0	48.7	50.8	54.3	58.3	59.9		
	10%	DB	75.0	71.8	69.4	63.6	58.4	54.1	53.1	54.8	58.6	63.3	67.8	71.0		
		MCWB	60.5	60.3	59.1	54.8	51.3	48.5	47.0	47.6	49.6	52.8	56.4	58.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	68.1	66.6	62.4	57.6	56.3	52.7	53.4	57.3	60.9	65.3	66.5		
		MCDB	83.7	76.2	76.7	69.3	60.6	58.1	55.9	58.8	65.2	66.5	73.7	79.2		
	2%	WB	66.9	65.9	64.5	60.0	55.3	52.4	50.2	51.4	54.2	57.9	61.9	64.0		
		MCDB	76.4	73.8	72.5	66.3	61.1	55.7	54.7	56.8	60.7	65.5	72.8	73.8		
	5%	WB	64.6	64.1	62.4	58.1	53.8	50.6	48.8	49.9	52.4	55.9	59.9	61.9		
		MCDB	74.9	71.8	69.7	64.6	59.1	54.4	53.6	55.4	59.0	64.6	69.5	70.7		
	10%	WB	62.4	62.2	60.4	56.0	52.0	49.2	47.5	48.3	50.7	53.8	57.9	59.8		
		MCDB	72.5	69.2	68.2	62.4	57.8	53.4	52.3	54.0	57.3	62.0	66.5	68.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.5	19.8	21.2	21.2	19.7	18.2	18.3	19.3	20.7	21.2	21.1	21.4		
		MCDBR	32.1	25.6	27.3	27.3	23.1	20.5	20.2	22.7	25.0	27.6	29.0	29.5		
		MCWBR	13.6	12.7	13.5	15.7	15.1	14.5	14.0	14.7	14.7	14.9	14.5	14.1		
	5% WB	MCDBR	25.1	21.0	22.2	22.4	19.2	17.5	17.9	19.3	21.4	22.9	24.5	24.0		
		MCWBR	12.8	12.0	11.9	13.9	13.5	13.3	13.2	13.3	13.8	13.9	13.9	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.335	0.329	0.320	0.310	0.294	0.288	0.285	0.300	0.315	0.323	0.329	0.331		
	taud		2.540	2.575	2.594	2.596	2.611	2.614	2.632	2.587	2.537	2.525	2.535	2.540		
	Ebn at Noon		315	308	298	279	263	253	264	279	294	307	315	318		
	Edh at Noon		34	31	28	24	20	18	19	23	29	32	34	34		
(44) All-Sky Solar Radiation	RadAvg		2139	1795	1417	965	636	498	565	824	1225	1638	1925	2165		
	RadStd		150	86	66	54	36	40	27	60	83	83	139	134		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.76	N/A	-1.51	+2.54	N/A	N/A	N/A	N/A	+227	+36
(47)	Regional (27 neighbors)	+0.59	N/A	N/A	+1.26	+0.58	N/A	N/A	-184	+196	+32

Nomenclature: See separate page